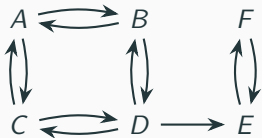


PageRank II

Hoa Vu

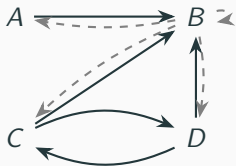
Fixing PageRank

Spider Traps



- The graph is strongly connected and has no dangling nodes.
- However, there is a spider trap: eventually, a random surfer will get stuck in E and F .
- The pagerank will be $1/2$ for E and F and 0 for the rest (which is not reasonable).

Fix 1: Dangling Nodes



- B is a sink: it has no outgoing edges.
- Fix: replace each sink with edges to *all* n nodes, each with weight $1/n$.
- Dashed edges show the added links from B .
- After the fix, every column of M sums to 1 (M is a proper stochastic matrix).

Fix 2: Teleportation

After handling sinks, there can still be spider traps and disconnected components.

Solution: allow the random surfer to jump to a page uniformly at random with probability ϵ (Google sets $\epsilon \approx 1/7$).

$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} + \epsilon \frac{1}{n}.$$

Revised Transition Matrix

The new transition matrix (e is the all-ones matrix):

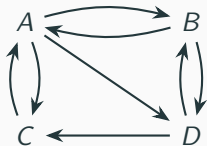
$$\hat{M} = (1 - \epsilon) M + \frac{\epsilon}{n} e.$$

$$\hat{M} = (1 - \epsilon) \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} + \epsilon \begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{bmatrix}.$$

To find the stationary distribution, solve $\pi = \hat{M}\pi$. In this case:

$$\pi_A = (1 - \epsilon)(\pi_B/2 + \pi_C) + \epsilon/4, \quad \pi_B = (1 - \epsilon)(\pi_A/3 + \pi_D/2) + \epsilon/4, \quad \dots$$

PageRank Example ($\epsilon = 0.12$)



With $\epsilon = 0.12$, $n = 4$, $\epsilon/n = 0.03$:

$$\pi_A = 0.88(\pi_B/2 + \pi_C) + 0.03$$

$$\pi_B = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_C = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_D = 0.88(\pi_A/3 + \pi_B/2) + 0.03$$

Solving: $\pi_A \approx 0.326$, $\pi_B \approx \pi_C \approx \pi_D \approx 0.224$.

Power Method

Computing PageRank: Power Method

- We compute PageRank iteratively (the power method).
- Start with $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$.
- For $t = 1, 2, \dots$, update $\pi^{(t)} = \hat{M}\pi^{(t-1)}$. Equivalently:

$$\pi_v^{(t)} = (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} + \epsilon \frac{1}{n}.$$

- The distribution $\pi^{(t)}$ converges to the stationary distribution π .
- Running time: $O(m + n)$ per iteration, where m is the number of edges.

Define the error at iteration t :

$$\text{error}(t) = \sum_v |\pi_v^{(t)} - \pi_v|.$$

Convergence Bound

$$\text{error}(t) \leq (1 - \epsilon)^t \cdot \text{error}(0).$$

- The error decreases by a factor of $(1 - \epsilon)$ each iteration.

Adjusted Computation: Stationary Equation

With the fix, a dangling node u contributes π_u/n to *every* page, while a normal node u contributes π_u/d_u to each neighbor.

The stationary equation $\pi = \hat{M}\pi$ becomes:

$$\pi_v = (1 - \epsilon) \left(\sum_{\substack{u \rightarrow v \\ d_u > 0}} \frac{\pi_u}{d_u} + \underbrace{\frac{1}{n} \sum_{\substack{u \\ d_u = 0}} \pi_u}_{\text{dangling mass}} \right) + \epsilon \frac{1}{n}.$$

Adjusted Computation: Power Iteration

The power-iteration update ($\pi^{(t)} = \hat{M}\pi^{(t-1)}$):

$$\pi_v^{(t)} = (1 - \epsilon) \left(\sum_{\substack{u \rightarrow v \\ d_u > 0}} \frac{\pi_u^{(t-1)}}{d_u} + \frac{1}{n} \sum_{\substack{u \\ d_u = 0}} \pi_u^{(t-1)} \right) + \epsilon \frac{1}{n}.$$

The dangling sum $\sum_{u: d_u=0} \pi_u^{(t-1)}$ is computed once per iteration, then added to every v .

PageRank Example: Power Iteration in Python

```
eps, n = 0.12, 4
base = eps / n
d = 1 - eps

A = B = C = D = 1 / n
for i in range(1, 11):
    A, B, C, D = (
        d * (B / 2 + C) + base,
        d * (A / 3 + D / 2) + base,
        d * (A / 3 + D / 2) + base,
        d * (A / 3 + B / 2) + base,
    )
    print(i, A, B, C, D)
```

PageRank Example: Power Iteration (1/2)

Starting from $\pi^{(0)} = (0.25, 0.25, 0.25, 0.25)$:

iter	π_A	π_B	π_C	π_D
0	0.2500	0.2500	0.2500	0.2500
1	0.3600	0.2133	0.2133	0.2133
2	0.3116	0.2295	0.2295	0.2295
3	0.3329	0.2224	0.2224	0.2224
4	0.3235	0.2255	0.2255	0.2255
5	0.3276	0.2241	0.2241	0.2241

PageRank Example: Power Iteration (2/2)

iter	π_A	π_B	π_C	π_D
6	0.3258	0.2247	0.2247	0.2247
7	0.3266	0.2245	0.2245	0.2245
8	0.3263	0.2246	0.2246	0.2246
9	0.3264	0.2245	0.2245	0.2245
10	0.3264	0.2245	0.2245	0.2245

Converges to the solved values: $\pi_A \approx 0.326$, $\pi_B \approx \pi_C \approx \pi_D \approx 0.224$.